B3: Quantum, Atomic & Molecular Physics — Model Solutions, Trinity 2015

Question 1 — Central field, LS coupling, 5s5l of Sr

Problem. Define central field approximation, configurations, terms, levels. Derive the Landé interval rule. Identify l in Sr 5s5l from three given level energies and assign L, S, J to all four levels. From six observed transition energies, identify l' in 5s5l' and assign L, S, J on both ends.

(a) Central field approximation, configurations, terms, levels

Replace the full electron–electron repulsion by a spherically symmetric one-body potential $U(r)$ (the spherical average of the nuclear attraction plus the mean field of the other electrons), and treat the residual

$$H_{\rm res} = \sum_{i<j} \frac{e^2}{4\pi\epsilon_0 r_{ij}} - \sum_i \big[U(r_i) + Ze^2/(4\pi\epsilon_0 r_i)\big]$$

as a perturbation.

In $U(r)$ each electron has good (n, l, m_l, m_s) and energies E_{nl}, so the zeroth-order eigenstate is fully specified by populations $(nl)^q$ — the *configuration*.

Residual Coulomb $H_{\rm res}$ commutes with total $\boldsymbol{L}$ and $\boldsymbol{S}$ but not with individual $\boldsymbol{l}_i, \boldsymbol{s}_i$:

$$\text{configuration} \longrightarrow \text{terms } {}^{2S+1}L, \qquad \text{splitting} \sim 10^4\,\text{cm}^{-1}.$$

Spin–orbit $H_{\rm SO} = \sum_i \xi(r_i)\boldsymbol{l}_i \cdot \boldsymbol{s}_i$ commutes only with total $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$:

$$\text{term} \longrightarrow \text{levels } {}^{2S+1}L_J, \qquad J = |L-S|, \ldots, L+S.$$

(b) LS coupling and the Landé interval rule

LS coupling: $H_{\rm res} \gg H_{\rm SO}$, so L, S are good quantum numbers and $H_{\rm SO}$ acts within a fixed ${}^{2S+1}L$ term.

Project $\sum_i \xi(r_i)\boldsymbol{l}_i \cdot \boldsymbol{s}_i$ onto the term: by the projection (Wigner–Eckart) theorem,

$$H_{\rm SO}\big|_{LS} = A_{LS}\, \boldsymbol{L} \cdot \boldsymbol{S}. \qquad (1)$$

Use $\boldsymbol{J} = \boldsymbol{L} + \boldsymbol{S}$:

$$\boldsymbol{L} \cdot \boldsymbol{S} = \tfrac{1}{2}(\boldsymbol{J}^2 - \boldsymbol{L}^2 - \boldsymbol{S}^2).$$

Eigenvalue in level J:

$$E_J = \tfrac{\hbar^2}{2} A_{LS} \big[J(J+1) - L(L+1) - S(S+1)\big].$$

Adjacent-level interval:

$$E_J - E_{J-1} = \tfrac{\hbar^2}{2} A_{LS} \big[J(J+1) - (J-1)J\big].$$

Simplify:

$$\boxed{E_J - E_{J-1} = \hbar^2 A_{LS}\, J} \tag{2}$$

The spacing between adjacent levels of a fine-structure multiplet is proportional to the larger J.

(c) Identifying l and assigning L, S, J for 5s5l

Configuration 5s5l has $s_1 = s_2 = \frac{1}{2}$, $l_1 = 0$, $l_2 = l$, hence $L = l$, $S = 0$ or 1:

$$\text{four levels: } {}^1L_l,\ {}^3L_{l-1},\ {}^3L_l,\ {}^3L_{l+1}.$$

The triplet is the only multiplet showing fine structure; the singlet is the isolated fourth level.

Triplet intervals from the data:

$$\Delta_1 = 35022 - 35007 = 15\ \text{cm}^{-1}, \qquad \Delta_2 = 35045 - 35022 = 23\ \text{cm}^{-1}.$$

Apply (2) with $J = L, L+1$ for the two adjacent gaps:

$$\frac{\Delta_2}{\Delta_1} = \frac{L+1}{L}.$$

Insert numbers:

$$\frac{23}{15} \simeq \frac{3}{2} \ \Rightarrow\ L = 2.$$

Hence $l = 2$ (5s5d), and the triplet is ${}^3\mathrm{D}_{1,2,3}$.

Assignments:

$$\boxed{\begin{array}{lll} 35\,007\ \text{cm}^{-1}: & {}^3\mathrm{D}_1: & L=2,\ S=1,\ J=1, \\ 35\,022\ \text{cm}^{-1}: & {}^3\mathrm{D}_2: & L=2,\ S=1,\ J=2, \\ 35\,045\ \text{cm}^{-1}: & {}^3\mathrm{D}_3: & L=2,\ S=1,\ J=3, \\ \text{fourth level}: & {}^1\mathrm{D}_2: & L=2,\ S=0,\ J=2. \end{array}}$$

(d) Identifying l' in 5s5l'

Six transitions $\Rightarrow$ a triplet–triplet multiplet with $\Delta L = \pm1, \Delta S = 0, \Delta J = 0, \pm1$ (no $0 \to 0$).

Trial $l' = 1$ (5s5p, ${}^3\mathrm{P}_{0,1,2}$): selection rules give

$${}^3\mathrm{D}_1 \to {}^3\mathrm{P}_{0,1,2}\ (3), \quad {}^3\mathrm{D}_2 \to {}^3\mathrm{P}_{1,2}\ (2), \quad {}^3\mathrm{D}_3 \to {}^3\mathrm{P}_2\ (1).$$

Total: six. Consistent.

Cluster the data by lower level. Three transitions to the same lower level differ by the upper-^{3}D intervals $15, 23$ cm^{-1}:

$$20\,108,\ 20\,123,\ 20\,146: \quad \text{intervals } 15, 23 \ \Rightarrow\ \text{common lower } {}^3\mathrm{P}_2.$$

Two transitions to the same lower level differ by the upper-${}^3\mathrm{D}_1 \to {}^3\mathrm{D}_2$ interval 15 cm^{-1}:

$$20\,503,\ 20\,518: \quad \text{interval } 15 \ \Rightarrow\ \text{common lower } {}^3\mathrm{P}_1.$$

Single transition:

$$20\,689: \quad \text{from } {}^3\mathrm{D}_1 \text{ only} \ \Rightarrow\ \text{lower } {}^3\mathrm{P}_0.$$

Lower-level energies:

$$E({}^3\mathrm{P}_2) = 35\,007 - 20\,108 = 14\,899\ \text{cm}^{-1},$$

$$E(^3\mathrm{P}_1) = 35\,007 - 20\,503 = 14\,504\ \mathrm{cm}^{-1},$$
$$E(^3\mathrm{P}_0) = 35\,007 - 20\,689 = 14\,318\ \mathrm{cm}^{-1}.$$

Landé check ($J = 1, 2$):

$$\frac{E_2 - E_1}{E_1 - E_0} = \frac{14\,899 - 14\,504}{14\,504 - 14\,318} = \frac{395}{186} \simeq 2,$$

matching $(L+1)/L = 2/1$ for $L = 1,\ S = 1$. Confirms ^{3}P.

$$\boxed{l' = 1\ \text{(5s5p)}, \qquad \text{lower } ^3\mathrm{P}_{0,1,2}:\ L = 1,\ S = 1,\ J = 0, 1, 2.}$$

Per-line assignments (upper → lower):

$$\boxed{\begin{array}{ll}
20\,108\ \mathrm{cm}^{-1}: & ^3\mathrm{D}_1\ (L, S, J = 2, 1, 1) \to {}^3\mathrm{P}_2\ (1, 1, 2), \\
20\,123\ \mathrm{cm}^{-1}: & ^3\mathrm{D}_2\ (2, 1, 2) \to {}^3\mathrm{P}_2\ (1, 1, 2), \\
20\,146\ \mathrm{cm}^{-1}: & ^3\mathrm{D}_3\ (2, 1, 3) \to {}^3\mathrm{P}_2\ (1, 1, 2), \\
20\,503\ \mathrm{cm}^{-1}: & ^3\mathrm{D}_1\ (2, 1, 1) \to {}^3\mathrm{P}_1\ (1, 1, 1), \\
20\,518\ \mathrm{cm}^{-1}: & ^3\mathrm{D}_2\ (2, 1, 2) \to {}^3\mathrm{P}_1\ (1, 1, 1), \\
20\,689\ \mathrm{cm}^{-1}: & ^3\mathrm{D}_1\ (2, 1, 1) \to {}^3\mathrm{P}_0\ (1, 1, 0).
\end{array}}$$

Question 2 — Hyperfine structure in a magnetic field, ^{39}K

Problem. Identify the three terms in the Hamiltonian and define weak/strong field. Show $\Delta E_F = g_F \mu_B B_{\mathrm{ext}} M_F$ in the weak limit and find g_F in terms of g_J. Compute g_J, F, g_F for ^{39}K 4s. Sketch the seven RF frequencies between hyperfine states and indicate which are visible along $\boldsymbol{B}$.

(a) Physical origin and field regimes

$A_J\, \boldsymbol{I}\cdot\boldsymbol{J}$: hyperfine interaction. The nuclear magnetic moment $\boldsymbol{\mu}_I = g_I \mu_N \boldsymbol{I}/\hbar$ couples to the magnetic field at the nucleus produced by the electrons; projection within a level gives $H_{\mathrm{hf}} = A_J \boldsymbol{I}\cdot\boldsymbol{J}$ (Fermi contact for s-electrons; orbital + dipolar for $l \neq 0$).

$g_J \mu_B \boldsymbol{J}\cdot B_{\mathrm{ext}}$: Zeeman energy of the electronic magnetic moment $\boldsymbol{\mu}_J = -g_J \mu_B \boldsymbol{J}/\hbar$ in the external field.

$-g_I \mu_N \boldsymbol{I}\cdot B_{\mathrm{ext}}$: Zeeman energy of the nuclear moment in the external field; smaller than the electronic Zeeman by $\mu_N/\mu_B \sim 10^{-3}$.

Compare the electronic Zeeman $g_J \mu_B B_{\mathrm{ext}}$ with the hyperfine spacing $\sim A_J |\boldsymbol{F}|$:

$$\text{weak: } g_J \mu_B B_{\mathrm{ext}} \ll A_J F \ \Rightarrow\ \boldsymbol{F} = \boldsymbol{I} + \boldsymbol{J} \text{ good},$$

$$\text{strong: } g_J \mu_B B_{\mathrm{ext}} \gg A_J F \ \Rightarrow\ M_I, M_J \text{ good (Paschen–Back of hfs).}$$

(b) Weak-field shift and g_F

Couple $\boldsymbol{F} = \boldsymbol{I} + \boldsymbol{J}$; $|F M_F\rangle$ diagonalises $A_J \boldsymbol{I}\cdot\boldsymbol{J}$. Treat the Zeeman term as a perturbation, dropping the small nuclear Zeeman. First-order shift:

$$\Delta E_F = \langle F M_F | g_J \mu_B \boldsymbol{J}\cdot B_{\mathrm{ext}} | F M_F \rangle.$$

Project $\boldsymbol{J}$ onto $\boldsymbol{F}$ (Wigner–Eckart within a fixed F):

$$\boldsymbol{J} \to \frac{\langle \boldsymbol{J}\cdot\boldsymbol{F} \rangle}{F(F+1)\hbar^2}\, \boldsymbol{F}.$$

Evaluate $\boldsymbol{J}\cdot\boldsymbol{F} = \frac{1}{2}(\boldsymbol{F}^2 + \boldsymbol{J}^2 - \boldsymbol{I}^2)$:

$$\langle \boldsymbol{J}\cdot\boldsymbol{F}\rangle = \tfrac{\hbar^2}{2}\big[F(F+1) + J(J+1) - I(I+1)\big].$$

Take $B_{\rm ext} = B_{\rm ext}\hat{\boldsymbol{z}}$, $\langle F_z\rangle = \hbar M_F$:

$$\Delta E_F = g_J \mu_{\rm B} B_{\rm ext} M_F \,\frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}.$$

Identify g_F:

$$\boxed{\Delta E_F = g_F \mu_{\rm B} B_{\rm ext} M_F, \qquad g_F = g_J\,\frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}.} \tag{3}$$

(c) ^{39}K 4s: g_J, F, g_F

Ground term 4s ^{2}S$_{1/2}$: $L = 0$, $S = \frac{1}{2}$, $J = \frac{1}{2}$. Landé formula:

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

Insert:

$$g_J = 1 + \frac{\frac{3}{4} + \frac{3}{4} - 0}{2\cdot\frac{3}{4}} = 1 + 1.$$

$$\boxed{g_J = 2.}$$

Couple $J = \frac{1}{2}$ with $I = \frac{3}{2}$:

$$F = |J - I|, \ldots, J + I = 1,\, 2.$$

Apply (3), $J(J+1) = \frac{3}{4}$, $I(I+1) = \frac{15}{4}$:
$F = 1$:

$$g_F = 2\cdot\frac{2 + \frac{3}{4} - \frac{15}{4}}{2\cdot 2} = 2\cdot\frac{-1}{4}.$$

$$\boxed{g_{F=1} = -\tfrac{1}{2}.}$$

$F = 2$:

$$g_F = 2\cdot\frac{6 + \frac{3}{4} - \frac{15}{4}}{2\cdot 6} = 2\cdot\frac{3}{12}.$$

$$\boxed{g_{F=2} = +\tfrac{1}{2}.}$$

(d) Seven RF frequencies between $F = 1$ and $F = 2$

Energy of a hyperfine sublevel (ΔF ignored as a constant offset between manifolds):

$$E(F, M_F) = E_0(F) + g_F \mu_{\rm B} B_{\rm ext} M_F.$$

RF transition frequency from $(F = 1, M)$ to $(F = 2, M')$:

$$h\nu = h\nu_0 + \mu_{\rm B} B_{\rm ext}\big(g_2 M' - g_1 M\big).$$

Substitute $g_1 = -\frac{1}{2}$, $g_2 = +\frac{1}{2}$:

$$h\nu = h\nu_0 + \tfrac{\mu_{\rm B} B_{\rm ext}}{2}\big(M' + M\big).$$

Magnetic-dipole selection: $\Delta F = \pm 1$, $\Delta M_F = 0, \pm 1$. Tabulate $(M + M')$:

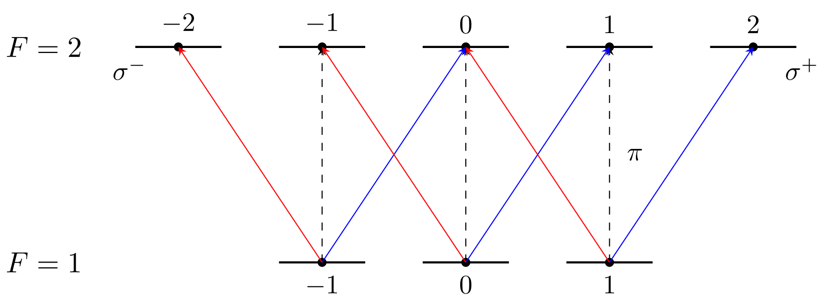
	$M=-1$	$M=0$	$M=+1$
$M' = M-1$ (σ^-)	–	-1	$+1$
$M' = M$ (π)	-2	0	$+2$
$M' = M+1$ (σ^+)	-1	$+1$	–

(missing entries: $M' = -2$ requires $M = -1 \to M' = -2$, giving $M + M' = -3$; $M' = +2$ requires $M = +1 \to M' = +2$, giving $M + M' = +3$).

Distinct values of $M + M'$:

$$\{-3,\ -2,\ -1,\ 0,\ +1,\ +2,\ +3\} \ \Rightarrow\ \boxed{\text{seven frequencies.}}$$

Longitudinal observation. Along $\boldsymbol{B}$, only $\sigma^\pm$ ($\Delta M_F = \pm 1$) radiation is emitted; π ($\Delta M_F = 0$) is forbidden. Visible shifts $(M + M') \in \{-3, -1, +1, +3\}$ from σ transitions only — four lines.



Solid (blue/red): $\sigma^\pm$ visible along $\boldsymbol{B}$. Dashed: π, not visible along $\boldsymbol{B}$.

Question 3 — ^{27}AlH band system, P/Q/R branches

Problem. Identify the terms in the rovibronic energy formula, give Λ, S for $X^1\Sigma^+$ and $A^1\Pi$, derive P, Q, R-branch frequencies, locate the band head and extract ν_{00}, $B'/(hc)$, equilibrium internuclear distances.

(a) Identifying the energy terms

$$E = E_e(R_0) + (n + \tfrac{1}{2})h\nu_v + B\,K(K+1).$$

$E_e(R_0)$**:** electronic (Born–Oppenheimer) energy at the equilibrium internuclear distance R_0.

$(n + \frac{1}{2})h\nu_v$**:** vibrational energy of small-amplitude harmonic motion about R_0, $\nu_v = \frac{1}{2\pi}\sqrt{k/\mu}$, μ reduced nuclear mass.

$B\,K(K+1)$**:** rigid-rotor energy at fixed R_0, $B = \hbar^2/(2\mu R_0^2)$, K the total nuclear-rotation quantum number.

(b) Quantum numbers of $X^1\Sigma^+$ and $A^1\Pi$

Term symbol $^{2S+1}\Lambda$ with $\Lambda = |L_z|$ (electronic orbital projection on the internuclear axis):

$$\boxed{X^1\Sigma^+ :\ S = 0,\ \Lambda = 0; \qquad A^1\Pi :\ S = 0,\ \Lambda = \hbar.}$$

(c) Branch frequencies

Transition energy (E' upper, E'' lower, $v' = v'' = 0$):

$$h\nu = h\nu_{00} + B'K'(K'+1) - B''K''(K''+1).$$

P branch ($\Delta K = +1$ in emission, $K'' = K'+1$, hence $K' = K''-1$):
Substitute $K' = K''-1$:

$$h\nu_P = h\nu_{00} + B'(K''-1)K'' - B''K''(K''+1).$$

Expand and collect powers of K'':

$$h\nu_P = h\nu_{00} + B'(K''^2 - K'') - B''(K''^2 + K'').$$

Group:

$$\boxed{h\nu_P = h\nu_{00} - (B'+B'')\,K'' + (B'-B'')\,K''^2.} \tag{4}$$

Q branch ($K' = K''$):

$$h\nu_Q = h\nu_{00} + (B'-B'')\,K''(K''+1).$$

$$\boxed{h\nu_Q = h\nu_{00} + (B'-B'')\,K'' + (B'-B'')\,K''^2.} \tag{5}$$

R branch ($K' = K''+1$):

$$h\nu_R = h\nu_{00} + B'(K''+1)(K''+2) - B''K''(K''+1).$$

Expand:

$$h\nu_R = h\nu_{00} + B'(K''^2 + 3K'' + 2) - B''(K''^2 + K'').$$

Group:

$$\boxed{h\nu_R = h\nu_{00} + 2B' + (3B'-B'')\,K'' + (B'-B'')\,K''^2.} \tag{6}$$

(d) Band head, ν_{00}, $B'/(hc)$, equilibrium separations

Treat P and R together via Fortrat parameter m: $m = -K''$ (P, $m \le -1$), $m = K''+1$ (R, $m \ge 1$). Both (4) and (6) reduce to

$$h\nu(m) = h\nu_{00} + (B'+B'')\,m + (B'-B'')\,m^2.$$

Differentiate w.r.t. m:

$$\frac{\mathrm{d}(h\nu)}{\mathrm{d}m} = (B'+B'') + 2(B'-B'')m.$$

Set to zero:

$$m_{\text{head}} = \frac{B'+B''}{2(B''-B')}.$$

For AlH the upper-state bond is longer, so $B' < B''$, giving $m_{\text{head}} > 0$: the head lies in the R branch. Translate $m_{\text{head}} = K''_{\text{head}} + 1$:

$$\boxed{K''_{\text{head}} = \frac{B'+B''}{2(B''-B')} - 1.} \tag{7}$$

Extracting $B'/(hc)$ from the Q branch. (5) written in wavenumbers ($\tilde{\nu} \equiv \nu/c$):

$$\tilde{\nu}_Q(K'') - \tilde{\nu}_{00} = (\tilde{B}' - \tilde{B}'')\,K''(K''+1).$$

Read from the figure: $\tilde{\nu}_Q(K'' = 22) \simeq 23\,200\ \text{cm}^{-1}$, and the $K'' \to 0$ extrapolation of the Q branch gives $\tilde{\nu}_{00} \simeq 23\,480\ \text{cm}^{-1}$ (this is also where Q meets the gap between R and P at low K''). Use $K''(K''+1) = 506$:

$$\tilde{B}' - \tilde{B}'' = \frac{23\,200 - 23\,480}{506}.$$

Numerator:

$$23\,200 - 23\,480 = -280.$$

Divide:

$$\tilde{B}' - \tilde{B}'' = \frac{-280}{506} = -0.553\ \text{cm}^{-1}.$$

Add $\tilde{B}'' = 6.3907\ \text{cm}^{-1}$:

$$\boxed{\tilde{B}' \equiv B'/(hc) = 5.84\ \text{cm}^{-1}, \qquad \tilde{\nu}_{00} \simeq 23\,480\ \text{cm}^{-1}.}$$

Consistency check via R-branch head. (7):

$$K''_{\text{head}} = \frac{5.84 + 6.39}{2(6.39 - 5.84)} - 1 = \frac{12.23}{1.10} - 1.$$

Evaluate:

$$K''_{\text{head}} \simeq 11.1 - 1 = 10.1.$$

Head wavenumber from (6) at $m_{\text{head}} = 11.1$:

$$\tilde{\nu}_{\text{head}} = \tilde{\nu}_{00} + (\tilde{B}' + \tilde{B}'')m_{\text{head}} + (\tilde{B}' - \tilde{B}'')m_{\text{head}}^2.$$

Insert numbers:

$$\tilde{\nu}_{\text{head}} = 23\,480 + 12.23 \cdot 11.1 - 0.553 \cdot 123.2.$$

$$\tilde{\nu}_{\text{head}} = 23\,480 + 135.8 - 68.1 \simeq 23\,548\ \text{cm}^{-1},$$

in agreement with the pile-up of R-branch points near $23\,545\ \text{cm}^{-1}$ in the figure.

Equilibrium internuclear distances. For ^{27}AlH:

$$\mu = \frac{27 \cdot 1}{28}\,\text{u} = 0.9643\,\text{u} = 1.6014 \times 10^{-27}\ \text{kg}.$$

$B = \hbar^2/(2\mu R_0^2) \Rightarrow B/(hc) = \hbar/(4\pi c\mu R_0^2)$, hence

$$R_0^2 = \frac{\hbar}{4\pi c\mu\,(B/(hc))}.$$

Numerator:

$$\hbar = 1.0546 \times 10^{-34}\ \text{J s}.$$

Denominator pieces:

$$4\pi c = 4\pi \cdot 2.998 \times 10^8 = 3.767 \times 10^9\ \text{m/s}.$$

$$4\pi c\mu = 3.767 \times 10^9 \cdot 1.6014 \times 10^{-27} = 6.033 \times 10^{-18}\ \text{kg m/s}.$$

Constant prefactor:

$$\frac{\hbar}{4\pi c\mu} = \frac{1.0546 \times 10^{-34}}{6.033 \times 10^{-18}} = 1.748 \times 10^{-17}\ \text{m}^2\,\text{m}^{-1}.$$

Convert $\tilde{B}'' = 6.3907\ \text{cm}^{-1} = 638.07\ \text{m}^{-1}$:

$$R_0''^2 = \frac{1.748 \times 10^{-17}}{638.07} = 2.740 \times 10^{-20}\ \text{m}^2.$$

$$R_0'' = 1.655 \times 10^{-10}\ \text{m}.$$

$$\boxed{R_0(X^1\Sigma^+) = 1.66\ \text{Å}.}$$

Convert $\tilde{B}' = 5.84\ \text{cm}^{-1} = 584\ \text{m}^{-1}$:

$$R_0'^2 = \frac{1.748 \times 10^{-17}}{584} = 2.993 \times 10^{-20}\ \text{m}^2.$$

$$R_0' = 1.730 \times 10^{-10}\ \text{m}.$$

$$\boxed{R_0(A^1\Pi) = 1.73\ \text{Å}.}$$

The bond is longer in the electronically excited $A^1\Pi$ state, consistent with the smaller B' inferred from the Q-branch slope and the appearance of the R-branch head.

Question 4 — X-ray transitions and K-edge filtering

Problem. Interpret the Moseley-type formula and the screening constants σ_n, σ_m. Define K_α and the K-edge; find σ_1, σ_2 for Ca. Derive the filter inequality and find the highest-Z element whose K_α is absorbed by element $Z - 1$.

(a) Form of the formula and the meaning of $\sigma_{n,m}$

The hydrogenic Bohr formula for an electron in shell n of nuclear charge Z is $E_n = -hcR_\infty Z^2/n^2$. In a multi-electron atom an inner-shell electron sees the nuclear charge partially screened by the other electrons:

$$Z \;\to\; Z_{\text{eff}}^{(n)} = Z - \sigma_n.$$

The transition energy between two such shells is the difference of effective hydrogenic levels:

$$E_{nm} = hcR_\infty \left[\frac{(Z-\sigma_n)^2}{n^2} - \frac{(Z-\sigma_m)^2}{m^2} \right]. \tag{8}$$

σ_n: effective screening of the nucleus by all other electrons, as seen by an electron in shell n during the transition. It is approximately equal to the number of electrons whose mean radius is smaller than that of shell n, but is reduced by exchange/correlation and includes a partial contribution from electrons in the same shell. σ_n depends on n but only weakly on Z (Moseley empirical rule).

(b) K_α and the K-edge; σ for Ca

K_α: radiative transition $L \to K$ following the production of a 1s vacancy ($n = 2 \to n = 1$, $\Delta l = \pm 1$). Using (8) with $n = 1$, $m = 2$:

$$E_{K_\alpha} = hcR_\infty \left[(Z-\sigma_1)^2 - \tfrac{1}{4}(Z-\sigma_2)^2 \right].$$

K-edge: minimum photon energy capable of ejecting a 1s electron to the continuum; the K electron's binding energy. From (8) with $m \to \infty$:

$$E_K = hcR_\infty (Z-\sigma_1)^2.$$

Calcium (Z=20). Use $hcR_\infty = 13.606$ eV.
K-edge:

$$(Z - \sigma_1)^2 = \frac{4039}{13.606}.$$

Divide:

$$(Z - \sigma_1)^2 = 296.85.$$

Square root:

$$Z - \sigma_1 = 17.229.$$

$$\boxed{\sigma_1 = 20 - 17.23 = 2.77.}$$

K_α:

$$\frac{E_{K_\alpha}}{hcR_\infty} = 296.85 - \tfrac{1}{4}(20 - \sigma_2)^2.$$

Insert $E_{K_\alpha}/(hcR_\infty) = 3692/13.606 = 271.41$:

$$\tfrac{1}{4}(20 - \sigma_2)^2 = 296.85 - 271.41 = 25.44.$$

Multiply by 4:

$$(20 - \sigma_2)^2 = 101.74.$$

Square root:

$$20 - \sigma_2 = 10.087.$$

$$\boxed{\sigma_2 = 9.91.}$$

Comment: $\sigma_1 \simeq 2.77$ exceeds the naive value 1 (one other K electron) because the operative screening for the K_α/K-edge process includes contributions from outer electrons reorganising around the inner-shell hole. $\sigma_2 \simeq 9.91$ is approximately the count of electrons inside the L shell plus a substantial fraction of the L electrons themselves — reflecting that L electrons screen each other strongly in addition to the K shell.

(c) Filter inequality

Efficient absorption requires the K_α photon of element Z to lie above the K-edge of element $Z - 1$:

$$E_{K_\alpha}(Z) > E_{K\text{-edge}}(Z - 1).$$

Cancel hcR_∞:

$$(Z - \sigma_1)^2 - \tfrac{1}{4}(Z - \sigma_2)^2 > (Z - 1 - \sigma_1)^2.$$

Let $a = Z - \sigma_1$, $b = Z - \sigma_2$:

$$a^2 - \tfrac{1}{4}b^2 > (a - 1)^2.$$

Expand the RHS:

$$a^2 - \tfrac{1}{4}b^2 > a^2 - 2a + 1.$$

Cancel a^2:

$$-\tfrac{1}{4}b^2 > -2a + 1.$$

Multiply by -4 (flip inequality):

$$b^2 < 8a - 4.$$

Substitute back a, b:

$$(Z - \sigma_2)^2 < 8(Z - \sigma_1) - 4.$$

Expand the LHS and rearrange:

$$Z^2 - (2\sigma_2 + 8)Z + (\sigma_2^2 + 8\sigma_1 + 4) < 0.$$

Solve the quadratic; take the larger root for the upper bound:

$$Z < \tfrac{(2\sigma_2+8)+\sqrt{(2\sigma_2+8)^2-4(\sigma_2^2+8\sigma_1+4)}}{2}.$$

Simplify by factoring 4 inside the root:

$$\boxed{Z < \sigma_2 + 4 + \sqrt{(\sigma_2 + 4)^2 - (\sigma_2^2 + 8\sigma_1 + 4)}.} \tag{9}$$

(d) Highest Z' and energy mismatch

Insert $\sigma_1 = 2.77$, $\sigma_2 = 9.91$.

Outer term:

$$\sigma_2 + 4 = 13.91.$$

Inside the root:

$$(\sigma_2 + 4)^2 - (\sigma_2^2 + 8\sigma_1 + 4).$$

Compute pieces:

$$(\sigma_2 + 4)^2 = 13.91^2 = 193.49,$$

$$\sigma_2^2 = 9.91^2 = 98.21, \quad 8\sigma_1 = 22.16,$$

$$\sigma_2^2 + 8\sigma_1 + 4 = 98.21 + 22.16 + 4 = 124.37.$$

Subtract:

$$193.49 - 124.37 = 69.12.$$

Square root:

$$\sqrt{69.12} = 8.314.$$

Total:

$$Z < 13.91 + 8.31 = 22.22.$$

Hence

$$\boxed{Z'_{\max} = 22 \text{ (Ti, filtered by Sc, } Z' - 1 = 21).}$$

Energy mismatch $\Delta E = E_{K_\alpha}(\mathrm{Ti}) - E_{K\text{-edge}}(\mathrm{Sc})$.

K-edge of Sc:

$$E_{K\text{-edge}}(21) = 13.606\,(21 - 2.77)^2.$$

Compute:

$$(21 - 2.77)^2 = 18.23^2 = 332.33.$$

$$E_{K\text{-edge}}(21) = 13.606 \cdot 332.33 = 4521 \text{ eV}.$$

K_α of Ti:

$$E_{K_\alpha}(22) = 13.606\big[(22 - 2.77)^2 - \tfrac{1}{4}(22 - 9.91)^2\big].$$

Pieces:

$$(22 - 2.77)^2 = 19.23^2 = 369.79,$$

$$(22 - 9.91)^2 = 12.09^2 = 146.17, \quad \tfrac{1}{4} \cdot 146.17 = 36.54.$$

Subtract:

$$369.79 - 36.54 = 333.25.$$

Multiply:

$$E_{K_\alpha}(22) = 13.606 \cdot 333.25 = 4534 \text{ eV}.$$

Difference:

$$\boxed{\Delta E = E_{K_\alpha}(\text{Ti}) - E_{K\text{-edge}}(\text{Sc}) \simeq 4534 - 4521 = 13 \text{ eV}.}$$

The K_α of Ti lies only ~ 13 eV above the K-edge of Sc; for elements with $Z \geq 23$ (V), the K_α photon falls below the K-edge of $Z - 1$ and the simple "Z filtered by $Z - 1$" rule fails.